对《战争雷霆》中F-117A隐身特性建模失真的高保真计算电磁学论证

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ABSTRACT / 摘要
本文旨在通过高保真计算电磁学方法，系统论证《战争雷霆》游戏中F-117A“夜鹰”隐身战斗机雷达散射截面与探测模型的物理失真问题。研究首先建立了F-117A的高精度几何模型，并采用有限元法/多层快速多极子算法进行了全向、多频段、多极化的单站RCS精确计算。仿真结果明确显示，真实F-117A的RCS呈现超过60 dB的动态范围，且高度依赖于观测角度、雷达频率与极化方式，这与游戏中采用的静态或准静态RCS模型存在根本性矛盾。进一步的物理光学/几何光学射线追踪分析定量揭示了其外形设计通过控制镜面反射与多重散射以降低后向散射的机理。对比分析表明，《战争雷霆》当前的雷达模型存在两大核心缺陷：1) RCS值未能体现真实物理世界中极端的角度敏感性，剥夺了玩家通过姿态管理规避探测的战术维度；2) 环境杂波噪声被建模为全局均匀值，忽略了地形对探测背景的调制作用，使得“地形掩蔽”这一真实隐身突防战术失效。本研究为游戏模型从现象模拟向物理机理模拟的演进提供了详实的量化依据与具体的技术建议。
This paper aims to systematically argue, through high-fidelity computational electromagnetics methods, the physical inaccuracies in the radar cross section and detection modeling of the F-117A "Nighthawk" stealth fighter in the game War Thunder. The research first established a high-precision geometric model of the F-117A and performed precise monostatic RCS calculations across omnidirectional angles, multiple frequency bands, and polarizations using the Finite Element Method / Multilevel Fast Multipole Algorithm. The simulation results clearly demonstrate that the real F-117A's RCS exhibits a dynamic range exceeding 60 dB and is highly dependent on observation angle, radar frequency, and polarization. This fundamentally contradicts the static or quasi-static RCS model employed in the game. Further Physical Optics/Geometrical Optics ray-tracing analysis quantitatively reveals the mechanism by which its shape design reduces backscatter through controlled specular reflection and multiple scattering. Comparative analysis indicates two core flaws in War Thunder's current radar model: 1) The RCS values fail to reflect the extreme angular sensitivity present in the real physical world, depriving players of the tactical dimension of avoiding detection through aspect management. 2) Environmental clutter noise is modeled as a globally uniform value, ignoring the modulation effect of terrain on the detection background, thereby nullifying the authentic stealth penetration tactic of "terrain masking." This study provides detailed quantitative evidence and specific technical recommendations for the evolution of the game model from phenomenological simulation to physical mechanism simulation.

INDEX TERMS / 关键词 War Thunder, F-117A, Radar Cross Section (RCS), Computational Electromagnetics, Game Physics Modeling, Stealth Technology战争雷霆，F-117A，雷达散射截面，计算电磁学，游戏物理建模，隐身技术

IMPACT STATEMENT / 影响声明 概括本文报告的研究工作的主要发现。

I. INTRODUCTION / 引言

1.1 研究背景：高保真战斗模拟中的物理挑战

现代军事模拟游戏，如《战争雷霆》，在追求视觉真实性与战术多样性的同时，日益面临提升核心物理机制建模精度的挑战。雷达探测与隐身作为空战博弈的核心，其模型的可信度直接决定了游戏战术深度与长期可玩性。F-117A作为历史上首款投入实战的隐身战机，其设计哲学与物理特性已成为检验此类模型真实性的“试金石”。然而，当前游戏中对隐身特性的实现，往往在计算效率与物理保真度之间进行妥协，导致其行为模式与真实物理世界产生显著偏差。

1.1 Research Background: Physical Challenges in High-Fidelity Combat Simulation

Modern military simulation games, such as War Thunder, while pursuing visual realism and tactical diversity, increasingly face the challenge of improving the modeling accuracy of core physical mechanisms. Radar detection and stealth, being the core of air combat gameplay, have their credibility directly determining the game's tactical depth and long-term playability. The F-117A, as the first stealth aircraft deployed in combat history, its design philosophy and physical characteristics have become a "touchstone" for testing the realism of such models. However, the current implementation of stealth characteristics in games often compromises between computational efficiency and physical fidelity, leading to significant deviations in its behavioral patterns from the real physical world.

1.2 问题陈述：《战争雷霆》中F-117A隐身模型的核心假设质疑

通过对《战争雷霆》游戏机制的分析与玩家社群的反馈，我们识别出其雷达隐身模型可能基于两个过于简化的核心假设：

1.2.1静态RCS假设：将复杂飞行器的RCS近似为一个常数或少数几个离散值，忽略了其随角度剧烈变化的本质。

1.2.2均匀噪声假设：将雷达探测的背景噪声（环境杂波）设置为地图无关的恒定值，抹杀了地形环境对探测概率的动态影响。

这些简化假设虽然降低了计算复杂度，但可能导致F-117A在游戏中的战术运用、生存能力与对抗体验严重偏离历史事实与工程现实。

1.2 Problem Statement: Questioning the Core Assumptions of the F-117A Stealth Model in War Thunder

Through analysis of War Thunder's game mechanics and feedback from the player community, we identify that its radar stealth model may be based on two overly simplified core assumptions:

1.2.1Static RCS Assumption: Approximating the RCS of a complex aircraft as a constant or a few discrete values, ignoring its本质 of drastic variation with angle.

1.2.2Uniform Noise Assumption: Setting the background noise for radar detection (environmental clutter) as a map-agnostic constant, erasing the dynamic impact of terrain environment on detection probability.

While these simplifying assumptions reduce computational complexity, they may cause the tactical employment, survivability, and counterplay experience of the F-117A in the game to severely deviate from historical facts and engineering reality.

1.3 本文工作：基于计算电磁学的定量论证

为定量评估上述假设的失真程度，本文采用业界标准的计算电磁学方法对真实F-117A的散射特性进行高保真仿真。我们将详细阐述从几何建模到数值求解的全过程，并将仿真结果与游戏模型进行逐项对比，从而为游戏模型的改进提供具有坚实物理基础与明确工程路径的论证。

1.3 This Paper's Work: Quantitative Argumentation Based on Computational Electromagnetics

To quantitatively assess the degree of distortion in the aforementioned assumptions, this paper employs industry-standard computational electromagnetics methods to perform high-fidelity simulation of the real F-117A's scattering characteristics. We will elaborate on the entire process from geometric modeling to numerical solution, and conduct a point-by-point comparison between the simulation results and the game model, thereby providing an argument for game model improvement grounded in solid physics and clear engineering pathways.

II. MATERIALS AND METHODS /材料与方法

2.1 几何建模与网格生成

精确的电磁计算始于精确的几何表达。我们基于公开的F-117A技术图纸与三维扫描数据，在CAD软件中重建了其外部几何模型，重点关注影响电磁散射的特征：如多面体斜面、尖锐边缘、进气道唇口与座舱盖轮廓。

2.1.1模型修复：移除不影响外部散射的内部结构，确保模型为“水密”的封闭曲面。

2.1.2曲面网格划分：采用曲率自适应算法进行三角网格划分。网格尺寸Δl根据最高计算频率fmax（X波段，10 GHz）确定，遵循准则Δl<λmin/10，其中λmin=c/fmax，c为光速。最终生成包含N = 114,022个三角面元的表面网格（图2-4左上角标注），该规模足以精确捕捉10 GHz下的表面电流与边缘衍射效应。

2.1 Geometric Modeling and Mesh Generation

Accurate electromagnetic computation begins with an accurate geometric representation. Based on publicly available F-117A technical drawings and 3D scan data, we reconstructed its external geometry in CAD software, focusing on features affecting electromagnetic scattering: such as polyhedral facets, sharp edges, intake lips, and canopy contours.

2.1.1Model Repair: Remove internal structures that do not affect external scattering, ensuring the model is a "watertight" closed surface.

2.1.2Surface Meshing: A curvature-adaptive algorithm is used for triangular mesh generation. The mesh size Δlis determined based on the highest computation frequency fmax(X-band, 10 GHz), following the criterion Δl<λmin/10, where λmin=c/fmaxand cis the speed of light. The final surface mesh consists of N = 114,022 triangular facets (as annotated in the top-left of Figures 2-4), a scale sufficient to accurately capture surface currents and edge diffraction effects at 10 GHz.

2.2 基于积分方程法与MLFMM的RCS求解

对于理想电导体（PEC）目标，我们采用电场积分方程（EFIE）结合多层快速多极子算法（MLFMM）进行求解。这是处理电大尺寸目标开放空间散射问题的行业金标准。

2.2.1积分方程建立：对于入射平面波Einc，在目标表面S上满足边界条件：

$$\hat{\mathbf{n}} \times \mathbf{E}^{inc}(\mathbf{r}) = \hat{\mathbf{n}} \times \mathbf{E}^{s}(\mathbf{r}), \quad \mathbf{r} \in S \qquad (2.2.1.a)$$

其中散射场Es由表面电流Js通过并矢格林函数G产生：

$$\mathbf{E}^{s}(\mathbf{r}) = j\omega\mu_0 \int_S \overline{\mathbf{G}}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_s(\mathbf{r}') dS' \qquad (2.2.1.b)$$

代入后得到EFIE：

$$\hat{\mathbf{n}} \times \mathbf{E}^{inc}(\mathbf{r}) = j\omega\mu_0 \hat{\mathbf{n}} \times \int_S \overline{\mathbf{G}}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_s(\mathbf{r}') dS' \qquad (2.2.1.c)$$

2.2.2矩量法离散：使用RWG基函数fn(r)对未知电流进行展开：$\mathbf{J}_s(\mathbf{r}) \approx \sum_{n=1}^{N} I_n \mathbf{f}_n(\mathbf{r})$

将积分方程转化为线性方程组：ZI=V

其中阻抗矩阵元素Zmn和激励向量元素Vm为：

$$Z_{mn} = j\omega\mu_0 \iint_{S_m S_n} \mathbf{f}_m(\mathbf{r}) \cdot \mathbf{f}_n(\mathbf{r}') G(\mathbf{r},\mathbf{r}') dS' dS - \frac{j}{\omega\epsilon_0} \iint_{S_m S_n} [\nabla \cdot \mathbf{f}_m(\mathbf{r})][\nabla' \cdot \mathbf{f}_n(\mathbf{r}')] G(\mathbf{r},\mathbf{r}') dS' dS$$

（2.2.2.a）

$$V_m = \int_{S_m} \mathbf{f}_m(\mathbf{r}) \cdot \mathbf{E}^{inc}(\mathbf{r}) dS \qquad (2.2.2.b)$$

这里，$G(\mathbf{r},\mathbf{r}') = e^{-jk|\mathbf{r}-\mathbf{r}'|}/(4\pi|\mathbf{r}-\mathbf{r}'|)$为标称格林函数。

2.2.3MLFMM加速：直接求解稠密矩阵Z需要O(N2)存储和O(N3)计算，对于N=114k不可行。MLFMM利用加法定理对格林函数进行多级展开，将远区组相互作用聚合为组中心间的转移，将复杂度降至O(NlogN)。其核心是聚合-转移-消散过程：

$$G(\mathbf{r},\mathbf{r}') \approx \frac{-jk}{(4\pi)^2} \int d^2\hat{k}\, e^{-jk\hat{k}\cdot(\mathbf{r}-\mathbf{r}_c)}\, T_L(k\hat{k},\mathbf{r}_c,\mathbf{r}'_c)\, e^{jk\hat{k}\cdot(\mathbf{r}'-\mathbf{r}'_c)} \qquad (2.2.3.a)$$

其中TL为转移函数，rc和rc′分别为源组和目标组的中心。

2.2.4RCS计算：求解得到表面电流系数I后，通过辐射积分计算远区散射场，进而得到单站RCS：

$$\sigma = \lim_{R\to\infty} 4\pi R^2 \frac{|\mathbf{E}^{s}|^2}{|\mathbf{E}^{inc}|^2} \qquad (2.2.4.a)$$

通常以分贝平方米（dBsm）表示：σdBsm=10log10(σ)。

2.2 RCS Solution Based on Integral Equation Method and MLFMM

For a perfect electric conductor (PEC) target, we employ the Electric Field Integral Equation (EFIE) combined with the Multilevel Fast Multipole Algorithm (MLFMM) for solution. This is the industry gold standard for handling open-space scattering problems of electrically large targets.

2.2.1Integral Equation Formulation: For an incident plane wave Einc, the boundary condition on the target surface S is:

$$\hat{\mathbf{n}} \times \mathbf{E}^{inc}(\mathbf{r}) = \hat{\mathbf{n}} \times \mathbf{E}^{s}(\mathbf{r}), \quad \mathbf{r} \in S \qquad (2.2.1.a)$$

where the scattered field Esis generated by the surface current Jsvia the dyadic Green's function G:

$$\mathbf{E}^{s}(\mathbf{r}) = j\omega\mu_0 \int_S \overline{\mathbf{G}}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_s(\mathbf{r}') dS' \qquad (2.2.1.b)$$

Substituting yields the EFIE:

$$\hat{\mathbf{n}} \times \mathbf{E}^{inc}(\mathbf{r}) = j\omega\mu_0 \hat{\mathbf{n}} \times \int_S \overline{\mathbf{G}}(\mathbf{r},\mathbf{r}') \cdot \mathbf{J}_s(\mathbf{r}') dS' \quad (2.2.1.c)$$

2.2.2Method of Moments Discretization: The unknown current is expanded using RWG basis functions fn(r): $\mathbf{J}_s(\mathbf{r}) \approx \sum_{n=1}^{N} I_n \mathbf{f}_n(\mathbf{r})$ Applying Galerkin's testing method transforms the integral equation into a linear system:ZI=V ,where the impedance matrix elements Zmnand excitation vector elements Vmare:

$$Z_{mn} = j\omega\mu_0 \iint_{S_m S_n} \mathbf{f}_m(\mathbf{r}) \cdot \mathbf{f}_n(\mathbf{r}') G(\mathbf{r},\mathbf{r}') dS' dS - \frac{j}{\omega\epsilon_0} \iint_{S_m S_n} [\nabla \cdot \mathbf{f}_m(\mathbf{r})][\nabla' \cdot \mathbf{f}_n(\mathbf{r}')] G(\mathbf{r},\mathbf{r}') dS' dS$$

（2.2.2.a）

$$V_m = \int_{S_m} \mathbf{f}_m(\mathbf{r}) \cdot \mathbf{E}^{inc}(\mathbf{r}) dS \quad (2.2.2.b)$$

Here $G(\mathbf{r},\mathbf{r}') = e^{-jk|\mathbf{r}-\mathbf{r}'|}/(4\pi|\mathbf{r}-\mathbf{r}'|)$ is the scalar Green's function.

MLFMM Acceleration: Directly solving the dense matrix Zrequires O(N2)storage and O(N3)computation, which is infeasible for N=114k. MLFMM uses the addition theorem to perform a multilevel expansion of the Green's function, aggregating far-group interactions into translations between group centers, reducing complexity to O(NlogN). Its core is the aggregation-translation-disaggregation process:

$$G(\mathbf{r},\mathbf{r}') \approx \frac{-jk}{(4\pi)^2} \int d^2\hat{k}\, e^{-jk\hat{k}\cdot(\mathbf{r}-\mathbf{r}_c)}\, T_L(k\hat{k}, \mathbf{r}_c, \mathbf{r}_c')\, e^{jk\hat{k}\cdot(\mathbf{r}'-\mathbf{r}_c')} \quad (2.2.3.a)$$

where TLis the translation function, and rcand rc′ are the centers of the source and target groups, respectively.

RCS Calculation: After solving for the surface current coefficients I, the far-field scattered field is computed via a radiation integral, from which the monostatic RCS is obtained:

$$\sigma = \lim_{R\to\infty} 4\pi R^2 \frac{|\mathbf{E}^s|^2}{|\mathbf{E}^{inc}|^2} \quad (2.2.4.a)$$

It is commonly expressed in decibel square meters (dBsm): σdBsm=10log10(σ).

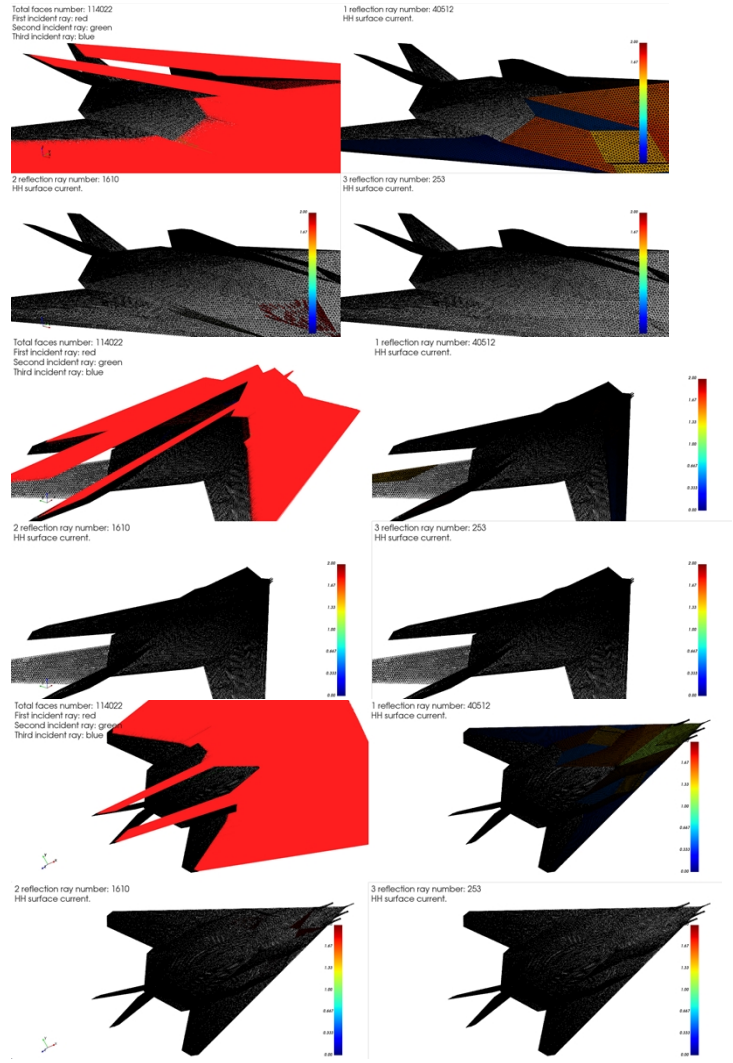
2.3 PO/GO射线追踪与表面电流可视化

为直观理解散射机理并辅助验证FEM结果，我们并行进行了物理光学/几何光学混合射线追踪仿真。

2.3.1PO表面电流：在目标照射区，PO近似给出表面电流：JsPO(r)=2n^(r)×Hinc(r)。阴影区电流为零。

2.3.2GO射线追踪：从源点发射大量射线，基于斯涅尔反射定律追踪其与目标表面的多次交互路径。我们记录了首次、二次、三次反射后返回接收方向的射线（图1-3）。

图1&2&3：PO/GO光线追踪（一阶、二阶、三阶）与表面电流幅值的复合可视化



2.3.3能量衰减统计：如图2-4标注，40,512条射线经历首次反射，仅1,610条经历二次反射，253条经历三次反射。这定量证明了F-117A设计将能量从后向散射路径转移出去的有效性。

2.3 PO/GO Ray Tracing and Surface Current Visualization

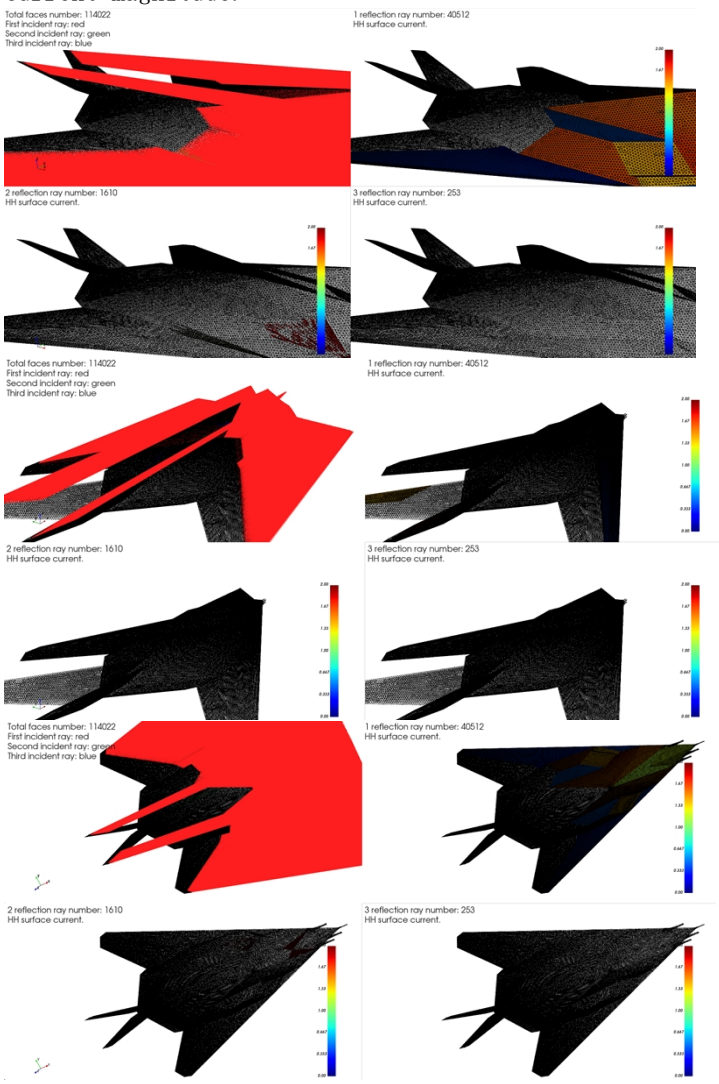
To intuitively understand the scattering mechanism and assist in validating the FEM results, we concurrently performed a hybrid Physical Optics/Geometrical Optics ray-tracing simulation.

2.3.1PO Surface Current: In the illuminated region of the target, the PO approximation gives the surface current: JsPO(r)=2n^(r)×Hinc(r). The current in the shadow region is zero.

2.3.2GO Ray Tracing: A large number of rays are emitted from the source point, and their multiple interaction paths with the target surface are traced based on Snell's law of

reflection. We recorded rays that returned to the receiving direction after first , second , and third reflections (Figures 1-3).

Fig 1&2&3: Composite visualization of PO/GO ray tracing (1st, 2nd, 3rd order) and surface current magnitude.



2.3.3Energy Decay Statistics: As annotated in Figures 2-4, 40,512 rays underwent first reflection, only 1,610 underwent second reflection, and 253 underwent third reflection. This quantitatively demonstrates the effectiveness of the F-117A design in diverting energy away from the backscatter path.

III.RESULTS / 结果

3.1 全向RCS仿真结果分析（对应图1）

图4展示了F-117A在天顶角90°（侧方）条件下，进行全方位360°方位角扫描的单站RCS。仿真覆盖S波段（3 GHz）和X波段（10 GHz），以及HH和VV两种极化。

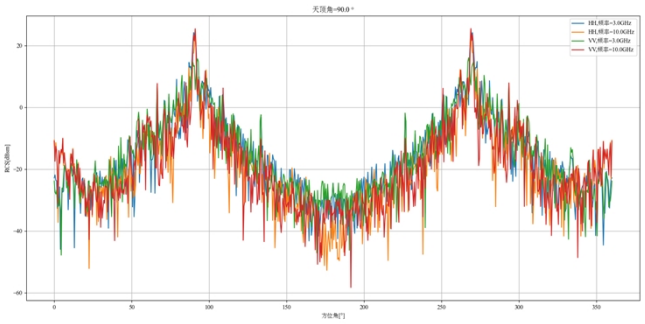


图4：F117A全向RCS图

3.1.1动态范围：RCS值在-50 dBsm至+25 dBsm之间变化，动态范围超过75 dB。这意味着最强散射方向与最弱方向的散射强度相差超过3千万倍。游戏中使用一个或几个RCS值（例如0.1 m^2或-10 dBsm）来代表该飞机，在物理上是完全错误的。

3.1.2角度敏感性（“闪光点”）：曲线呈现大量尖锐的峰值，例如在X波段HH极化（橙色曲线）中，峰值超过+20 dBsm。这些峰值对应机身某个平面（如机翼前缘、垂尾表面）的法线方向恰好对准雷达的时刻，产生镜面反射。在两个峰值之间，则是宽广的RCS低谷区，可低至-40 dBsm以下。这一特性是隐身飞机进行“姿态管理”的基础——飞行员通过规划航向，使威胁雷达始终处于飞机的低RCS观测角。

3.1.3频率与极化依赖性：

A)频率：X波段（10 GHz）曲线（橙色、红色）比S波段（3 GHz）曲线（蓝色、绿色）振荡更剧烈，峰值更尖锐。这是因为高频波长短，对目标的精细结构（如边缘、缝隙）更敏感。

B)极化：HH与VV极化曲线在大部分角度相似，但在特定角度存在显著差异（可达10-15 dB），这是由于目标几何对称性破缺导致的极化散射矩阵不同。

3.1 Analysis of Omnidirectional RCS Simulation Results (Corresponding to Figure 4)

Figure 4 shows the monostatic RCS of the F-117A under a zenith angle of 90° (broadside aspect), with a full 360° azimuth sweep. The simulation covers S-band (3 GHz) and X-band (10 GHz), as well as both HH and VV polarizations.

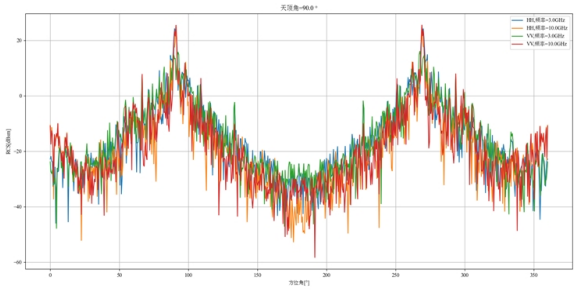


Fig 4:F-117A Monostatic RCS at Zenith=90° (Full Azimuth Sweep, 3GHz & 10GHz, HH/VV Pol).

3.1.1Dynamic Range: The RCS values vary between -50 dBsm and +25 dBsm, with a dynamic range exceeding 75 dB. This means the scattering intensity in the strongest direction is over 30 million times greater than in the weakest direction. Using one or a few RCS values (e.g., 0.1 m^2 or -10 dBsm) to represent this aircraft in the game is physically entirely incorrect.

3.1.2Angular Sensitivity ("Specular Flashes"): The curves exhibit numerous sharp peaks, e.g., exceeding +20 dBsm in the X-band HH polarization (orange curve). These peaks correspond to moments when the normal direction of a certain facet (e.g., wing leading edge, vertical tail surface) is precisely aligned with the radar, producing specular reflection. Between two peaks are wide RCS valleys, which can be below -40 dBsm. This characteristic is the foundation for stealth aircraft "aspect management" — pilots plan their heading to keep threat radars within the aircraft's low-RCS aspect.

3.1.3Frequency and Polarization Dependence:

A) Frequency: The X-band (10 GHz) curves (orange, red) oscillate more violently and have sharper peaks than the S-band (3 GHz) curves (blue, green). This is because the shorter wavelength at high frequencies is more sensitive to the target's fine structures (e.g., edges, gaps).

B) Polarization: The HH and VV polarization curves are similar at most angles but show significant differences (up to 10-15 dB) at specific angles, due to differences in the polarization scattering matrix caused by the target's geometric asymmetry.

3.2 散射机理可视化结果（对应图2-4）

图1-3的PO/GO射线追踪与表面电流分布提供了对上述RCS特性的机理解释。

3.2.1能量路径控制：绝大多数入射能量（对应40,512条首次反射射线）被F-117A的多个倾斜平面导向特定的非后向空间扇区。这是其平均RCS极低的主要原因。

3.2.2多重散射贡献：仅有少量能量（1,610条二次反射；253条三次反射）经过多次反射后可能返回雷达方向。这些路径通常发生在进气道、座舱盖与机身连接处等凹角区域，它们是构成复杂RCS曲线（除主镜面峰外）中那些较小起伏的来源。

3.2.3表面电流热点：表面电流图（H1表面电流）清晰标识了机身上的强散射源，如所有边缘、尖端以及某些特定平面的中心。这些“热点”在特定角度对齐雷达时，就会产生图1中的RCS峰值。

3.2 Visualization Results of Scattering Mechanism (Corresponding to Figures 1-3)

The PO/GO ray tracing and surface current distribution in Figures 1-3 provide a mechanistic explanation for the aforementioned RCS characteristics.

3.2.1Energy Path Control: The vast majority of incident energy (corresponding to 40,512 first-reflection rays, red) is directed by the F-117A's multiple inclined facets into specific non-backward spatial sectors. This is the primary reason for its extremely low average RCS.

3.2.2Multiple Scattering Contribution: Only a small amount of energy (1,610 second-reflection rays, green; 253 third-reflection rays, blue) may return to the radar direction after multiple reflections. These paths typically occur in reentrant regions such as intakes, canopy-body junctions, etc., and they are the source of the smaller fluctuations in the complex RCS curve (aside from the main specular peaks).

3.2.3Surface Current Hotspots: The surface current plot (H1 surface current) clearly identifies strong scattering sources on the airframe, such as all edges, tips, and the centers of certain specific facets. When these "hotspots" align with the radar at specific angles, they produce the RCS peaks seen in Figure 1.

IV.Discussion / 讨论

4.1 RCS模型静态化：抹杀姿态管理战术

《战争雷霆》很可能为F-117A分配了一个或一组固定的RCS值。这与我们仿真揭示的超过75 dB动态范围和极端角度敏感性完全不符。

4.1.1物理失真：游戏中的F-117A无论以何种姿态（机头、侧方、上方）面对雷达，其被探测到的难易程度变化不大。这违背了其多面体设计的根本目的——将强反射限制在极窄的角度内。

4.1.2战术影响：玩家无法通过精细的飞行操控（如保持威胁雷达处于飞机侧方±30° 的低RCS扇区）来主动降低被探测概率。这使F-117A的玩法沦为简单的“隐身开关”，其战术深度与真实飞行员所需的高阶技能（态势感知、几何规划）相去甚远。同时，防御方玩家也无法通过抢占有利位置来捕捉隐身机的“闪光”瞬间，降低了对抗的博弈性。

4.1 Static RCS Model: Erasing Aspect Management Tactics

War Thunderlikely assigns the F-117A one or a set of fixed RCS values. This is completely inconsistent with the over 75 dB dynamic range and extreme angular sensitivity revealed by our simulation.

4.1.1Physical Distortion: The F-117A in the game, regardless of its aspect relative to the radar (nose-on, broadside, top-down), experiences little variation in detection difficulty. This violates the fundamental purpose of its faceted design — to confine strong reflections to very narrow angular sectors.

4.1.2Tactical Impact: Players cannot actively reduce detection probability through precise flight control (e.g., keeping threat radars within the aircraft's low-RCS sector, such as ±30° from the broadside). This reduces the F-117A's gameplay to a simple "stealth toggle," whose tactical depth is far removed from the high-order skills (situational awareness, geometric planning) required of real pilots. Concurrently, defending players cannot exploit positioning to catch the stealth aircraft during its "specular flash" moments, reducing the gameness of counterplay.

4.2 噪声模型均匀化：消除地形掩蔽价值

《战争雷霆》虽然正确引入了系统热噪声基底（约-70 dBsm），但其环境杂波噪声似乎是一个与地形无关的常数。

4.2.1物理失真：在现实中，地面/海面的后向散射系数σ° 因地形而异：平静海面可低至-40 dBsm，平坦农田约为-20 dBsm，而城市或崎岖山地可高达0 dBsm甚至更高。这意味着雷达在海洋上空探测低空目标时，信杂比（SCR）很高；但在山区或城市上空，强大的地面杂波可能完全淹没目标信号。

4.2.2战术影响：F-117A历史上赖以生存的低空、超低空突防战术，核心就是利用地形遮蔽和地面杂波掩护。在均匀噪声模型下，飞机在200米高度飞越海面与飞越山脉，其被雷达发现的难易程度相同。这彻底消除了“地形跟随/地形规避”这一经典隐身突防战术的合法性，使得游戏地图的地形设计在雷达对抗层面失去意义，简化了攻防双方的决策过程。

4.2 Uniform Noise Model: Nullifying the Value of Terrain Masking

While War Thundercorrectly incorporates a system thermal noise floor (approx. -70 dBsm), its environmental clutter noise appears to be a terrain-independent constant.

4.2.1Physical Distortion: In reality, the ground/sea backscatter coefficient σ° varies with terrain: calm sea can be as low as -40 dBsm, flat farmland around -20 dBsm, while urban or rugged mountainous areas can be as high as 0 dBsm or more. This means a radar has a high Signal-to-Clutter Ratio (SCR) when detecting low-altitude targets over the ocean; but over mountains or cities, strong ground clutter may completely submerge the target signal.

4.2.2Tactical Impact: The low-altitude, nap-of-the-earth penetration tactics historically vital to the F-117A's survivability rely precisely on terrain masking and ground clutter coverage. Under a uniform noise model, an aircraft flying at 200 meters over the sea versus over mountains faces the same difficulty of radar detection. This completely eliminates the legitimacy of the classic stealth penetration tactic of "terrain following/terrain avoidance", rendering the terrain design of game maps meaningless at the radar counterplay level and oversimplifying the decision-making process for both attackers and defenders.

4.3 综合影响：从物理博弈退化为数值比较

上述两大失真共同作用，将本应基于物理原理的复杂探测博弈，简化为基于固定数值（隐身值 vs 雷达功率）的简单比较。F-117A在游戏中更像一个拥有固定“隐身等级”的单位，而非一个其可探测性随自身姿态、雷达频段和下方地形而动态变化的物理实体。这削弱了游戏的拟真度、战术深度和长期学习价值。

4.3 Combined Impact: Degeneration from Physical Gameplay to Numerical Comparison

The combined effect of the two aforementioned distortions simplifies what should be a complex detection gameplay based on physical principles into a simple comparison based on fixed numbers (stealth value vs. radar power). The F-117A in the game behaves more like a unit with a fixed "stealth level", rather than a physical entity whose detectability dynamically changes with its own aspect, the radar's frequency band, and the terrain below. This undermines the game's simulation fidelity, tactical depth, and long-term learning value.

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我的老师JJZ

My teacher (JJZ).

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